

NILOY SAHA

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EDUCATION

Hofstra University

B.S. in Computer Science

New York, USA

Expected Graduation, May 2027

- o **GPA:** 3.46/4.00, Dean's List
- o **Related Coursework:** Data Structures & Algorithms, Computer Organization & Programming, Software Engineering, Object-Oriented Programming, Applied NLP and Text Mining

EXPERIENCE

Student Technician

Maurice A. Deane School of Law at Hofstra University

New York, USA

Sept 2025 - Current

- **Cut troubleshooting time by 15%** by developing Python scripts to automate recurring IT support tasks, reinforcing practical programming skills.
- **Ensured 99% uptime** for several law school devices by configuring and monitoring networks using **TCP/IP and DNS protocols**.
- **Reduced repeat support tickets by 20%** through building a structured knowledge base, leveraging database organization and information retrieval principles.

PROJECTS

Smart Syllabus Calendar Sync | [GitHub](#) | [Site](#)

- **Engineered** a Next.js 14 application that **automatically transformed academic syllabi into interactive calendars**, reducing manual event entry time for students by **80%** through PDF parsing, date extraction, and AI classification.
- **Improved cross-platform accessibility for all users** by integrating **Google OAuth2 and ICS export** functionality, enabling seamless synchronization with **Google Calendar**.
- **Increased user engagement by 40%** through optimizing the calendar UI with **React Big Calendar and Tailwind CSS**, delivering a mobile-responsive interface with color-coded events and multiple viewing modes.

CourseCrafter AI | [GitHub](#) | [Site](#)

- **Built** a full-stack **Node.js & Express** web app that auto-generates structured AI learning courses from user-provided topics, streamlining course creation.
- Reduced **API response times by 30%** by integrating OpenAI API and YouTube Data API with Axios and optimizing backend performance in a Node.js & Express application.
- **Designed** a modern, responsive UI using **EJS templating and Bootstrap**, ensuring **seamless**, accessible learning experiences across devices.

Credit Card Fraudulent Predictor | [GitHub](#)

- **Engineered** a fraud detection system in Python using scikit-learn on **50k+ transactions**, applying data preprocessing, normalization, and SMOTE to address class imbalance
- Achieved **99%+ ROC-AUC** and **0.99 F1-score** by training and optimizing Random Forest and Gradient Boosting classifiers, enabling real-time fraud prediction with high precision.
- **Visualized** key patterns in transaction data with Matplotlib, improving **model interpretability** and driving insights into feature importance and risk factors.

ACTIVITIES AND LEADERSHIP

HOFSTRA C.O.D.E

Member

New York, USA

Sept 2024 - Current

- Placed top 5 in a weekly hackathon with a team of 4.
- Engaged in peer-led workshops to explore emerging technologies and share practical insights with peers.
- Contributed to open-source projects and group coding initiatives, applying learned concepts to real-world scenarios.

SKILLS

Programming Language: Java, Python, JavaScript, HTML/CSS, SQL, Dart, TypeScript

Frameworks and Concepts: NodeJs, ReactJs, Natural Language Processing, Machine Learning, Spring Boot, Bootstrap, Data Structures and Algorithms, Object-Oriented Programming, SDLC, RESTful APIs, Git/GitHub